# PHASE SUMMATION PATTERNS AND CENTROID GEOMETRY OF RIEMANN ZETA ZEROS

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Abstract. We investigate a phase representation of the nontrivial zeros of the Riemann zeta function based on the cumulative process $S_n = \sum_{k \le n} e^{i\gamma_k}$, where $\gamma_n$ denotes the imaginary part of the $n$-th zero. Using approximately two million zeros from Odlyzko's database, we study the geometry of the resulting complex trajectory and its associated phase-correlation structure. The cumulative trajectory exhibits a highly stable nonzero centroid, while the angular distribution remains essentially maximally uniform. Since the phases $e^{i\gamma_n}$ are known to be equidistributed on the unit circle, these observations suggest a subtle collective property of the ordered sequence of zeros. After recentering, the radial distribution displays a persistent bimodal structure whose detailed shape evolves slowly with sample size but whose two-scale character remains stable throughout the dataset. To quantify the underlying phase organization, we introduce correlation moments derived from the phase-correlation function $\rho(m) = \langle e^{i(\gamma_{n+m} - \gamma_n)} \rangle$. Numerically, the moment $M_0$ appears compatible with convergence toward 1, while the weighted moment $M_1$ appears compatible with convergence toward $-1/2$. If confirmed, these limiting values would indicate the presence of nontrivial long-range phase correlations extending across multiple scales. The work is therefore intended as an exploratory investigation identifying new observables.

## 1. Introduction

The statistical behaviour of the nontrivial zeros of the Riemann zeta function exhibits deep connections with random matrix theory.

Since Montgomery's pair-correlation conjecture [1] and the large-scale computations of Odlyzko [2], the local statistics of the zeros have repeatedly shown remarkable agreement with the Gaussian Unitary Ensemble (GUE).

Most investigations focus on normalized spacings, pair-correlation functions, nearest-neighbour distributions, or spectral form factors.

The present work adopts a different viewpoint.

Instead of studying normalized gaps directly, we consider the phases

$$e^{i\gamma_n} \tag{1}$$

and investigate the geometry generated by their cumulative summation.

This construction produces a complex trajectory whose properties can be studied using tools from geometry, harmonic analysis, and statistical physics.

The most striking empirical feature is the existence of a highly stable nonzero centroid.

Motivated by this observation, we introduce a phase-correlation framework that allows the geometric observations to be reformulated in terms of quantities admitting a more direct mathematical interpretation.

## 2. Data and Methodology

All computations were performed using Odlyzko's public database of Riemann zeta zeros containing N = 2,001,052 nontrivial zeros.

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The stored values possess finite decimal precision, approximately ten decimal places in the version used for the present computations.

If

$$\tilde{\gamma}_n = \gamma_n + \varepsilon_n, \qquad |\varepsilon_n| \lesssim 10^{-10} \tag{2}$$

then

$$e^{i\tilde{\gamma}_n} = e^{i\gamma_n}\left(1 + i\varepsilon_n + O(\varepsilon_n^2)\right) \tag{3}$$

so the induced perturbation on the phase process is of order $10^{-10}$.

Since

$$N^{-1/2} \approx 7 \times 10^{-4} \tag{4}$$

numerical rounding effects remain several orders of magnitude below the intrinsic statistical scale and are therefore regarded as negligible for the present study.

The central geometric object is the cumulative trajectory

$$S_n = \sum_{k \le n} e^{i\gamma_k} \tag{5}$$

Its centroid is defined by

$$C = \frac{1}{N}\sum_{n=1}^{N} S_n \tag{6}$$

To analyse the trajectory independently of its global translation, we also consider the centred process

$$S_n - C \tag{7}$$

The resulting radial and angular observables form the basis of the geometric analysis developed below.

## 3. Results

The cumulative phase process (5) generates a complex trajectory exhibiting a highly non-random large-scale structure.

Visual inspection immediately reveals that the trajectory does not behave like a standard two-dimensional random walk.

Instead, the cloud of points appears displaced from the origin and organized around a remarkably stable geometric centre.

This observation motivated the introduction of the centroid observable and the subsequent statistical analysis of its stability.

The dataset was divided into 200 independent blocks.

For each block a local centroid $C_k$ was computed.

The global centroid obtained from the full dataset is

$$C \approx 0.776907 + 0.767845\,i \tag{8}$$

corresponding to a modulus

$$|C| \approx 1.0923 \tag{9}$$

The radial structure of the centred trajectory exhibits a persistent bimodal distribution.

The radial distribution is not stationary in the strict sense, since its detailed shape evolves slowly with the number of accumulated zeros. Nevertheless, the presence of two dominant radial components persists over the full range of investigated sample sizes.

The data therefore suggest a slowly drifting but structurally stable bimodal radial law.

Figure 1 shows the density of the cumulative trajectory $S_n$ in the complex plane.

The cloud is approximately isotropic in angle, consistent with the known equidistribution of the phases. At the same time, the density is clearly not concentrated around the origin. Instead, it occupies an annular region whose geometric centre remains displaced from zero.

Figure 2 displays the corresponding radial density after centering. The distribution exhibits a pronounced bimodal structure, revealing the presence of two preferred radial scales. Although the precise location of the peaks changes slowly as larger portions of the dataset are included, the bimodal character persists and appears remarkably robust.

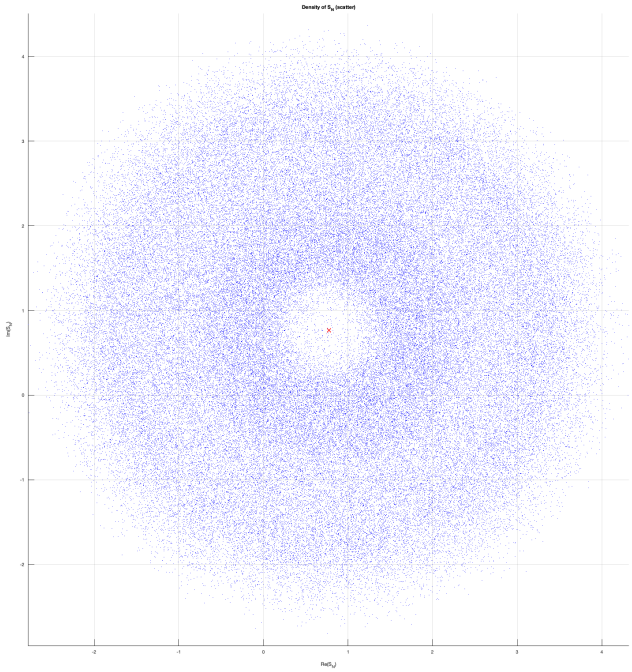


FIGURE 1. Density cloud of the cumulative phase trajectory $S_n$ in the complex plane. The red marker indicates the centroid. The cloud is approximately angularly uniform but remains significantly displaced from the origin.

The resulting cloud of block centroids clusters tightly around this value and exhibits remarkable stability.

Numerically,

$$\mathrm{std}(|C_k|) \approx 0.00174 \tag{10}$$

and

$$\mathrm{std}(\arg C_k) \approx 0.00158 \tag{11}$$

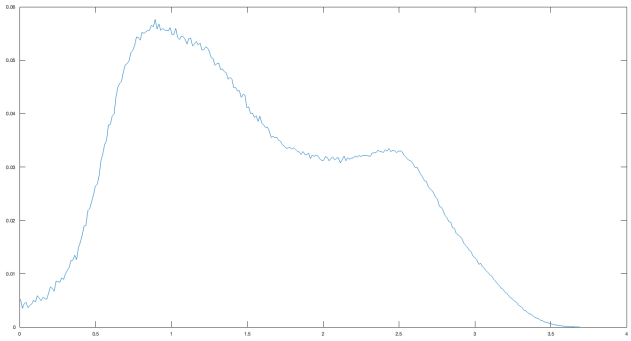


FIGURE 2. Empirical radial density of the centered trajectory. A persistent bimodal structure is observed. The positions of the maxima drift slowly with increasing sample size $N$, but the bimodal character remains stable throughout the dataset.

Relative fluctuations therefore remain below one percent, demonstrating that both the modulus and the direction of the centroid are highly stable across independent blocks of zeta zeros.

The stability persists throughout the entire dataset and strongly suggests that the observed centroid is not a finite-sample artefact.

To investigate the geometry independently of the global displacement, the trajectory was recentered according to

$$S_n \mapsto S_n - C \tag{12}$$

The angular distribution of the recentered trajectory was then analysed.

The measured angular entropy was

$$H = 5.192913 \tag{13}$$

whereas

$$\log(180) = 5.192957 \tag{14}$$

The difference lies well below the level expected from finite sampling.

Consequently, the angular distribution is essentially maximally uniform.

The coexistence of a highly stable nonzero centroid, an almost perfectly uniform angular distribution, and a persistent bimodal radial density constitutes one of the most unusual empirical features observed in the phase trajectory. These geometric observations naturally motivate the search for statistical quantities capable of quantifying the underlying phase organization. While the centroid, angular entropy, and radial density capture geometric aspects of the trajectory, they do not directly quantify the phase relationships responsible for these structures.

For this reason it is natural to introduce correlation observables capable of probing long-range phase coherence. To this end we introduce the phase-correlation moments

$$M_0(L) = \sum_{m \le L} \operatorname{Re} \rho(m), \tag{15}$$

and

$$M_1(L) = \sum_{m \le L} m \operatorname{Re} \rho(m), \tag{16}$$

where the correlation function $\rho(m)$ will be defined in the next subsection.

The quantity $M_0(L)$ measures the cumulative balance of phase correlations across all lags up to $L$, whereas $M_1(L)$ gives additional weight to long-range correlations.

Empirically, $M_0(L)$ exhibits a remarkable degree of stability. For cutoffs ranging from a few hundred to several thousand lags, the partial sums remain confined to a narrow interval and oscillate around the value – 0.5 with slowly decreasing amplitude.

The numerical evidence therefore suggests the asymptotic behaviour

$$\lim_{L\to\infty} M_0(L) = -\frac{1}{2}, \tag{17}$$

although no theoretical explanation is presently known.

The first moment $M_1(L)$ displays an apparent trend towards 1, as a structured behaviour. First experiments with surrogates indicates that the observed phenomenon is not restricted to a simple excess of short-range correlations, but involves a coherent contribution from a broad range of scales.

## 4. Conclusions

The phase representation considered in this work reveals a collection of empirical features that appear remarkably stable across more than two million nontrivial zeros of the Riemann zeta function.

The first and perhaps most unexpected observation is the existence of a nonzero centroid in the cumulative phase trajectory. At first sight, the presence of a preferred direction in the complex plane might appear difficult to reconcile with the quidistributed phases. However, equidistribution concerns the asymptotic angular distribution of individual phases, whereas the centroid probes cumulative coherence within the ordered sequence of zeros. The observed angular entropy remains essentially maximal, indicating that the phases are indeed distributed in an almost uniform manner. The centroid therefore suggests the existence of a subtle global organization that is compatible with equidistribution and cannot be detected through angular statistics alone.

A second notable feature is the persistent bimodal structure of the radial distribution. Although the precise location of the radial maxima evolves slowly with increasing sample size, the existence of two dominant radial scales remains stable throughout the dataset. Such behaviour is difficult to reconcile with the simplest random-walk picture, for which one would expect a single characteristic radial scale. The bimodality therefore points toward the existence of at least two competing geometric regimes governing the cumulative phase process. Whether this phenomenon is a direct consequence of known zero correlations or reflects a previously unstudied aspect of the phase representation remains unclear.

The correlation moments provide a complementary statistical description of these geometric observations. Numerically, the first weighted moment appears compatible with convergence toward (-1/2). If this behaviour persists in larger datasets and admits a theoretical explanation, it would imply a highly nontrivial cancellation mechanism among phase correlations extending over a broad range of scales. Such a value is not suggested by naive independence assumptions and would therefore encode genuine information about the collective organization of the zeros.

At the same time, the zeroth weighted moment appears compatible with convergence toward (1). Interpreted at a heuristic level, this would indicate the persistence of a net positive contribution from the cumulative phase-correlation structure. In combination with the behaviour of the first moment, this suggests that the correlation landscape is not dominated by purely local effects but instead possesses a coherent large-scale component. The simultaneous appearance of these two limiting values may therefore reflect a balance between constructive and destructive contributions distributed across different correlation lengths.

None of these observations presently contradict the established GUE description of local zero statistics. Rather, they suggest that the phase representation may provide access to global collective

features that are largely invisible when the zeros are studied only through normalized spacings or pair-correlation functions.

The results presented here should therefore be regarded primarily as exploratory. Their principal contribution is not the establishment of new theorems, but the identification of concrete empirical structures and quantitative observables that may serve as targets for future analytical investigation. Determining whether these phenomena follow from existing GUE theory, represent previously unnoticed consequences of known correlation laws, or encode genuinely new information about the zeta zeros remains an open problem.

## References


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